

Implementing Dataloaders

batch	label	Text
0	ham	...
1	ham	...
2	spam	...
3	ham	...
4	ham	...

Text messages are of varying length

To batch these messages, we have 2 options

(1) Truncate all messages to length of shortest message

(2) Pad all messages to length of longest message

BPE tokenizer

Input text

"First message"

1) Tokenize tokens

817, 2410

2) Pad to longest sequence

717, 2410, 50256, 50256,

end of text token

↙ Padded token IDs

we use 50256 as padding token

"another test message"

1194, 2420, 3275

50256, ..., 50256

1194, 2420, 3275,

50256, ..., 50256

"<end of text>" : 50256

"This is the third test message, which is very long"

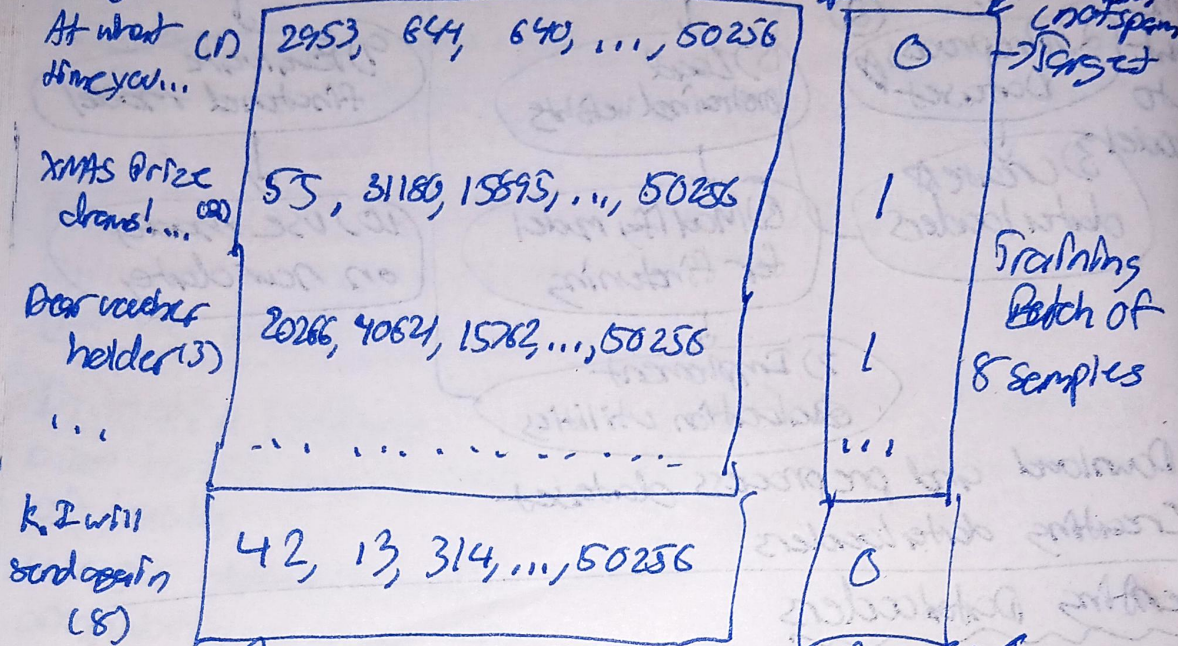
1212, 318, 262, 2368, 2420, 3275, 11, 543, 318, 845, 890

→ same as before

no padding in this case b/c it's longest message

Procedure!

Input text $\rightarrow$ Tokenize $\rightarrow$ Ensure that all sentences are same length
 original message padded to 120 tokens
 where 1 stands for "spam" and 0 for "not spam"



Each batch consists of 8 training examples

Each row (row) represents the token IDs corresponding to the original message text

Class label of 8th training example

247 "spam"

247 "not spam"

1494 total samples

1045 in training = 1045

[8, 120]

[130, 8, 120]

[38, 8, 120]

[19, 8, 120]

train-loader dimensions

test-loader dims.

val-loader dims.

8 samples $\rightarrow$ # of batches = $\frac{1045}{8} \approx 130$ total training batches